

Dynamometers

by

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Dynamometers: a dynamometer is a brake but in addition, it has a device to measure frictional resistance.

- **Absorption Dynamometers :**

1. Prony brake dynamometer
2. Rope brake dynamometer

- **Transmission Dynamometers**

1. Belt transmission dynamometer
2. Epicyclic dynamometer
3. Torsion dynamometer

Absorption Dynamometers

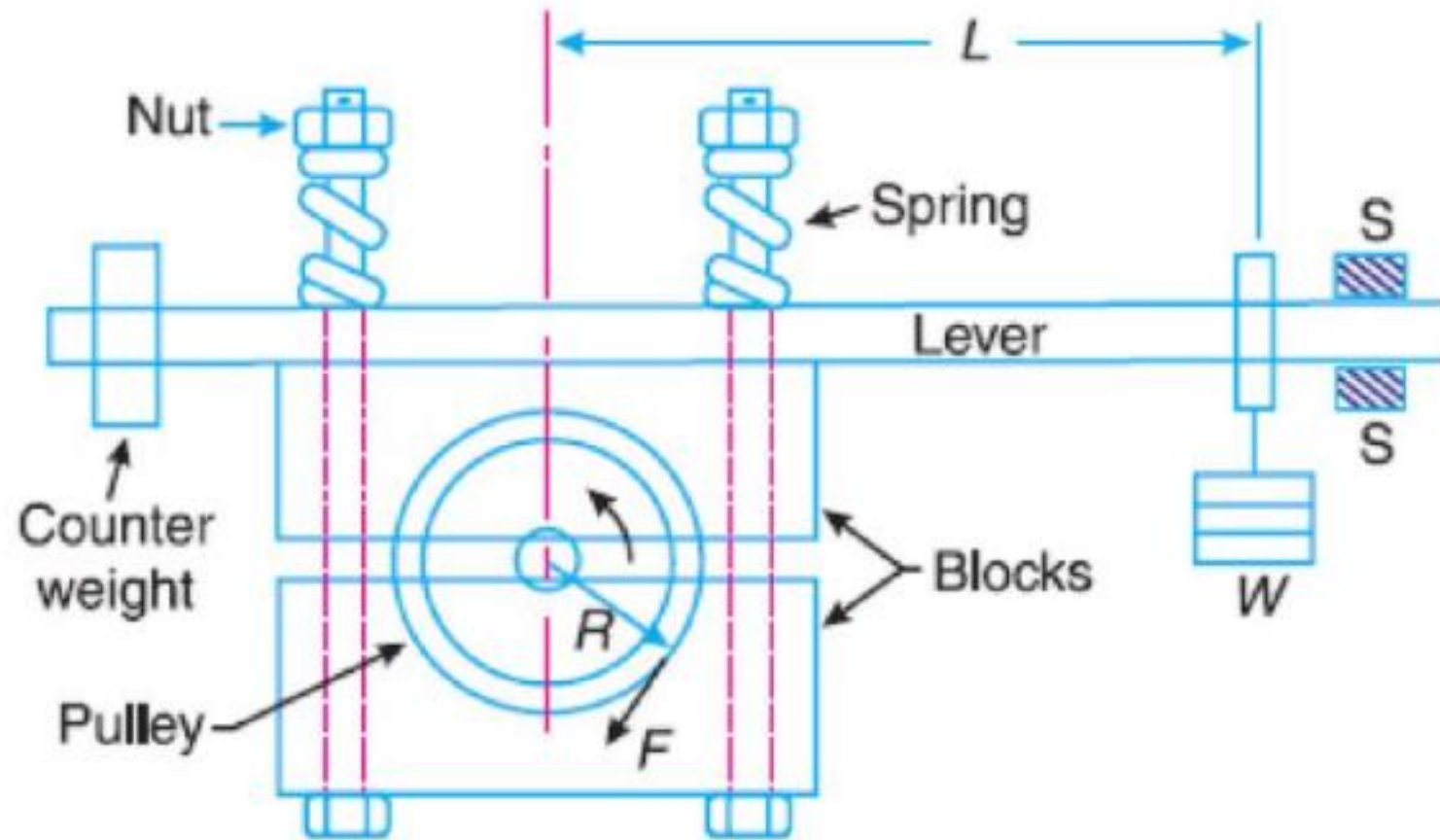
In the Absorption dynamometers, the entire energy or power produced by the engine is absorbed by the friction resistances of the brake and it is transformed into heat, during the process of measurement.

Used for measurement of moderate powers only.

Transmission Dynamometer

In the Transmission dynamometers, the energy is not wasted in friction but is used for doing work. The energy or power produced by the engine is transmitted through dynamometer to some other machines where power developed is suitably measured.

Prony brake dynamometer:



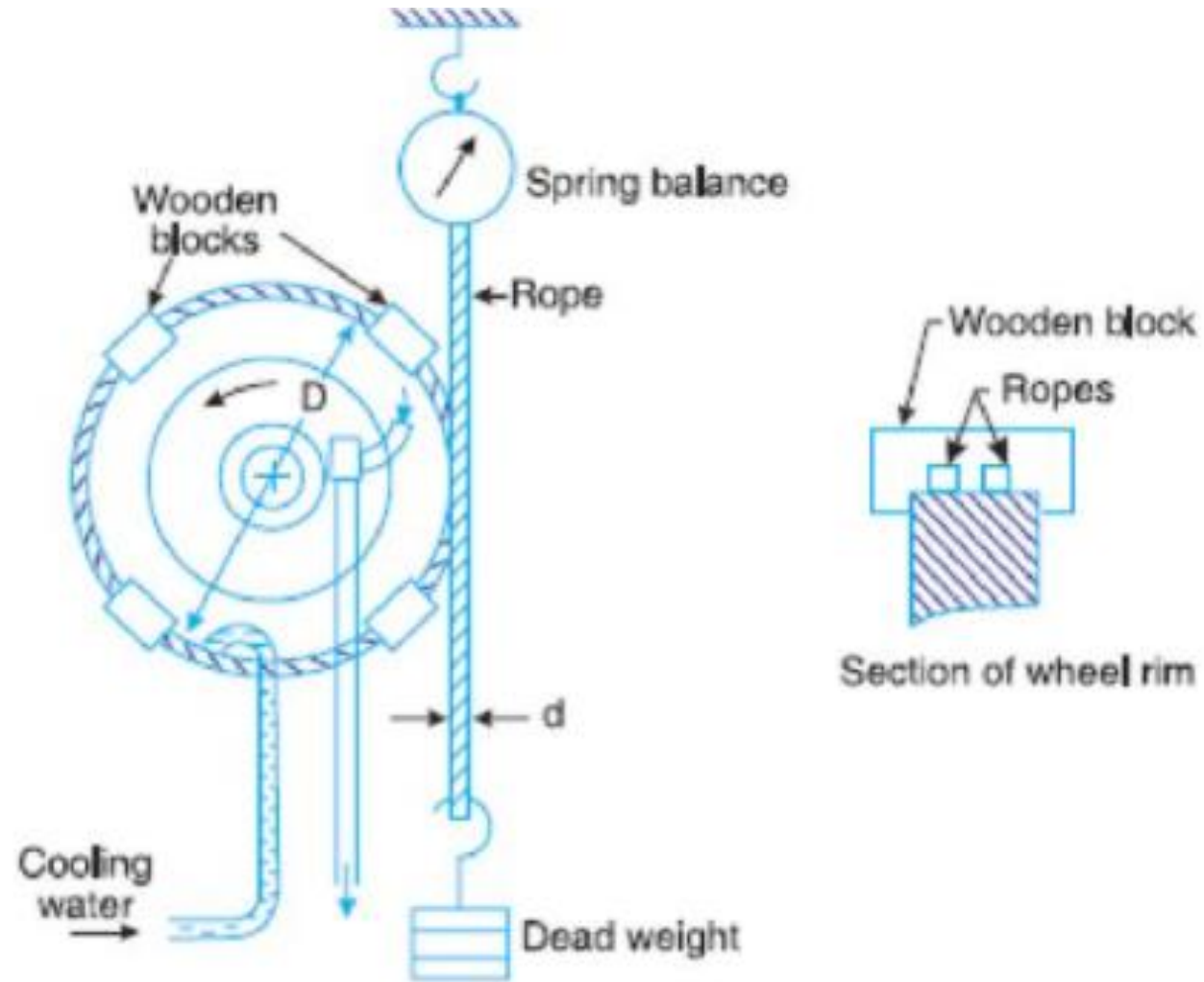
Prony brake dynamometer:

Moment of the frictional resistance: $T_f = WL = FR$

Work done per minute: $= 2\pi NT_f$

Brake power: $B.P. = \frac{2\pi NT_f}{60 \times 1000} = \frac{2\pi NWL}{60 \times 1000} \text{ kW}$

Rope brake dynamometer: it consists of one or more ropes wound around the flywheel or rim of the pulley, fixed rigidly to the shaft of the prime mover.



Rope brake dynamometer:

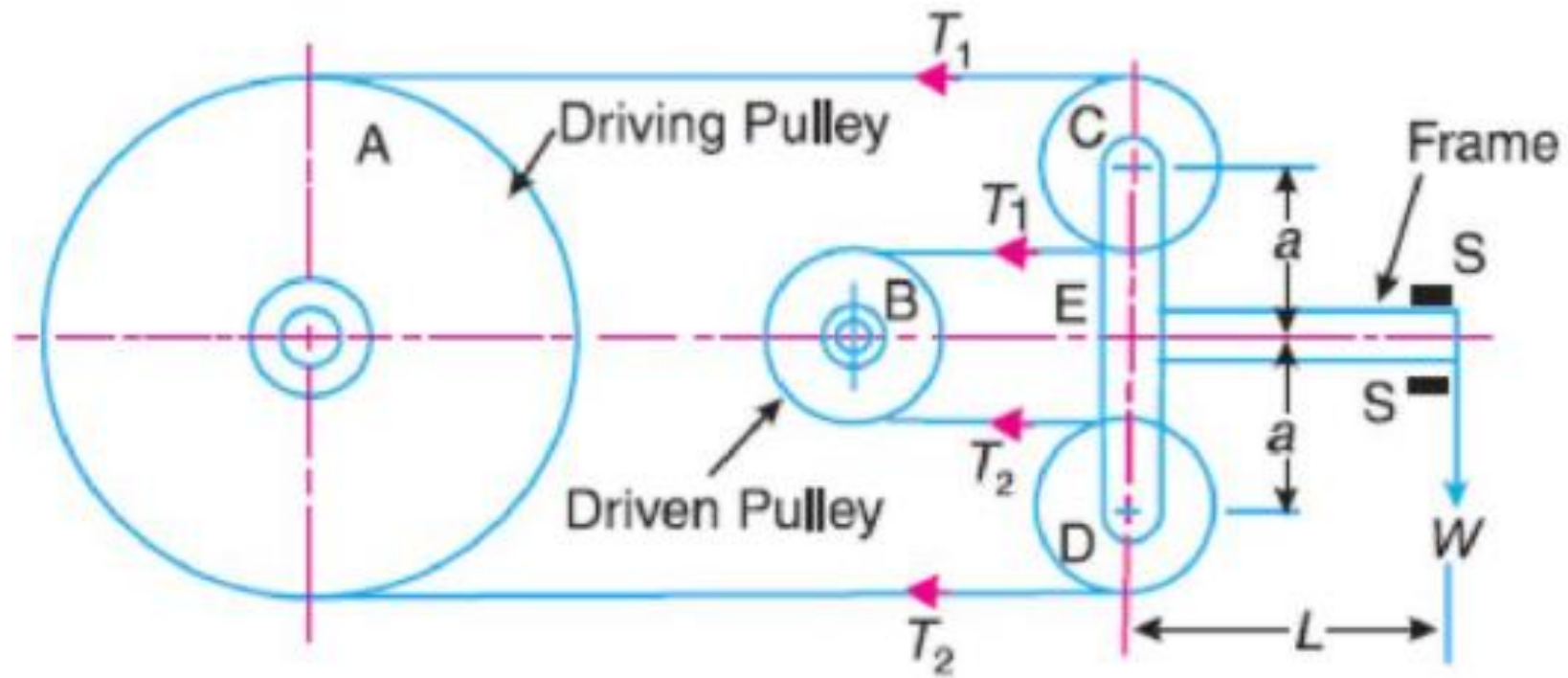
Work done per minute: $= (W - S) \pi (D + d) N$

Here, W is dead load on the rope, S is spring balance reading,

D is the diameter of pulley, d is diameter of rope, N is speed of the pulley.

Brake Power: $= \frac{(W - S) \pi (D + d) N}{60 \times 1000} \text{ kW}$

Belt transmission / Froude / Thorneycroft dynamometer:



Belt transmission dynamometer:

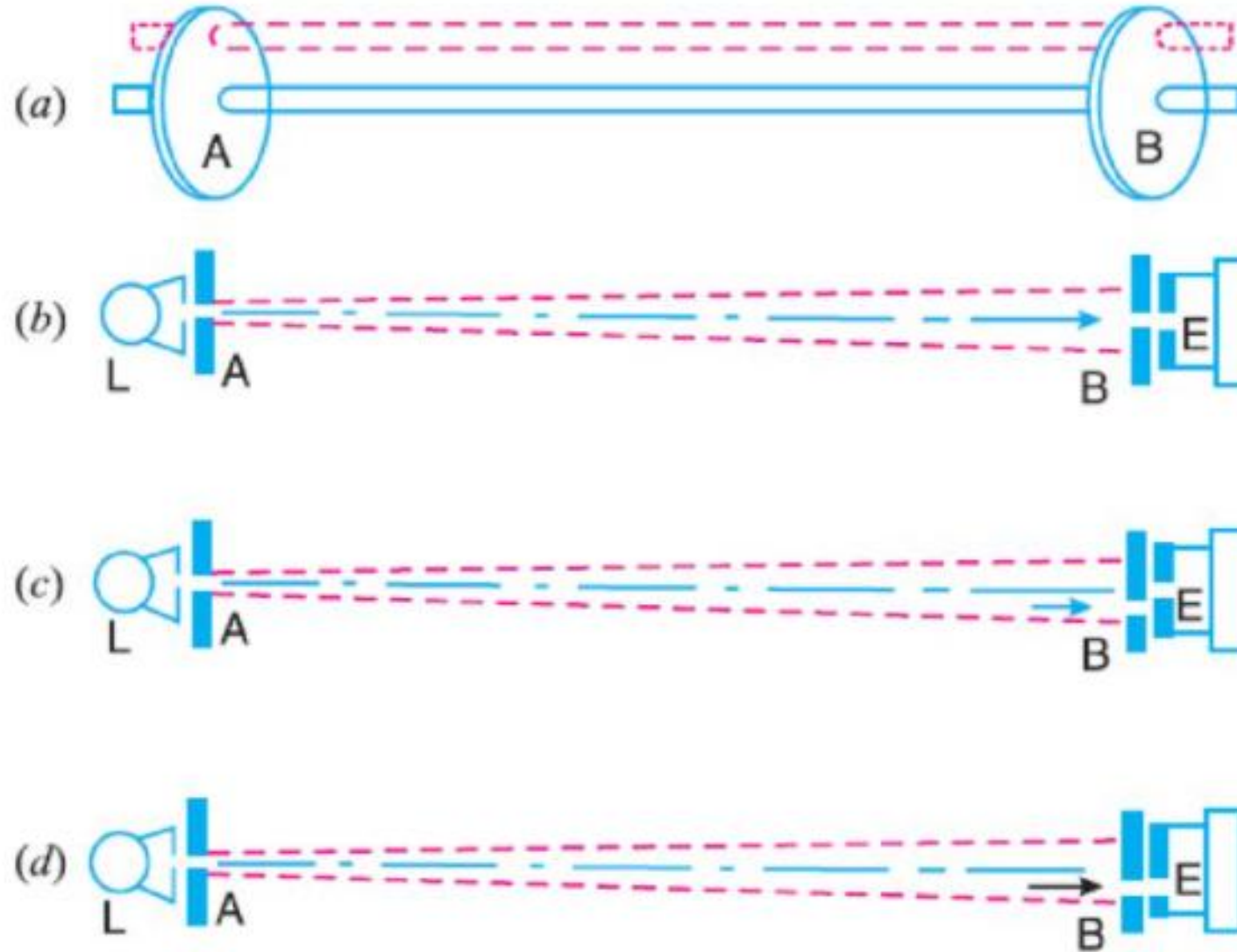
Taking moment about fulcrum E: $2T_1a = 2T_2a + WL$

$$T_1 - T_2 = \frac{WL}{2a}$$

Brake Power Developed:

$$B.P. = \frac{\pi DN(T_1 - T_2)}{60 \times 1000} \text{ kW}$$

Torsion dynamometer / Bevis Gibson Flash Light dynamometer:



Torsion dynamometer: is used to measure large power developed by a turbine or marine engines.

The torque developed by a shaft of diameter d , length L , and modulus of rigidity G is:

$$T = \left(\frac{GJ}{l} \right) \theta = k\theta$$

Here θ is angle of twist of the shaft, J polar moment of inertia of the shaft.

For Solid shaft:

$$J = \frac{\pi D^4}{32}$$

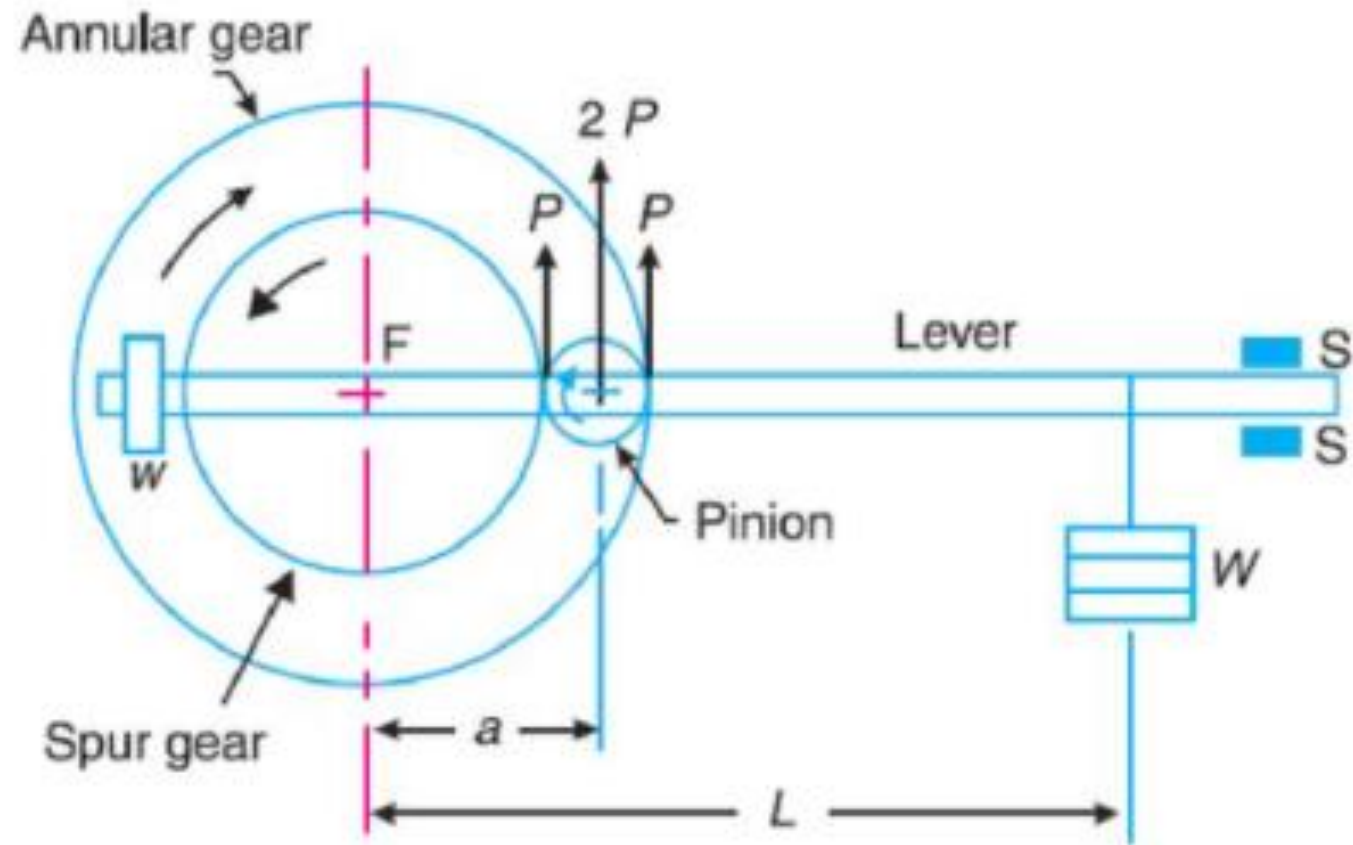
For Hollow shaft:

$$J = \frac{\pi(D^4 - d^4)}{32}$$

Power developed:

$$P = \frac{2\pi NT}{60 \times 1000} \text{ kW}$$

Epicyclic Train dynamometer :



Epicyclic Train dynamometer :

For the equilibrium of the lever, taking moments about the fulcrum:

$$2P \times a = W.L$$

Or

$$P = \frac{W.L}{2a}$$

Torque transmitted: $T = P.R$

Here, R is pitch circle radius of the spur gear

Power transmitted: $B.P. = \frac{2\pi NT}{60 \times 1000} = \frac{2\pi NPR}{60 \times 1000} kW$

Example 19.18. The essential features of a transmission dynamometer are shown in Fig. 19.35. *A* is the driving pulley which runs at 600 r.p.m. *B* and *C* are jockey pulleys mounted on a horizontal beam pivoted at *D*, about which point the complete beam is balanced when at rest. *E* is the driven pulley and all portions of the belt between the pulleys are vertical. *A*, *B* and *C* are each 300 mm diameter and the thickness and weight of the belt are neglected. The length *DF* is 750 mm.

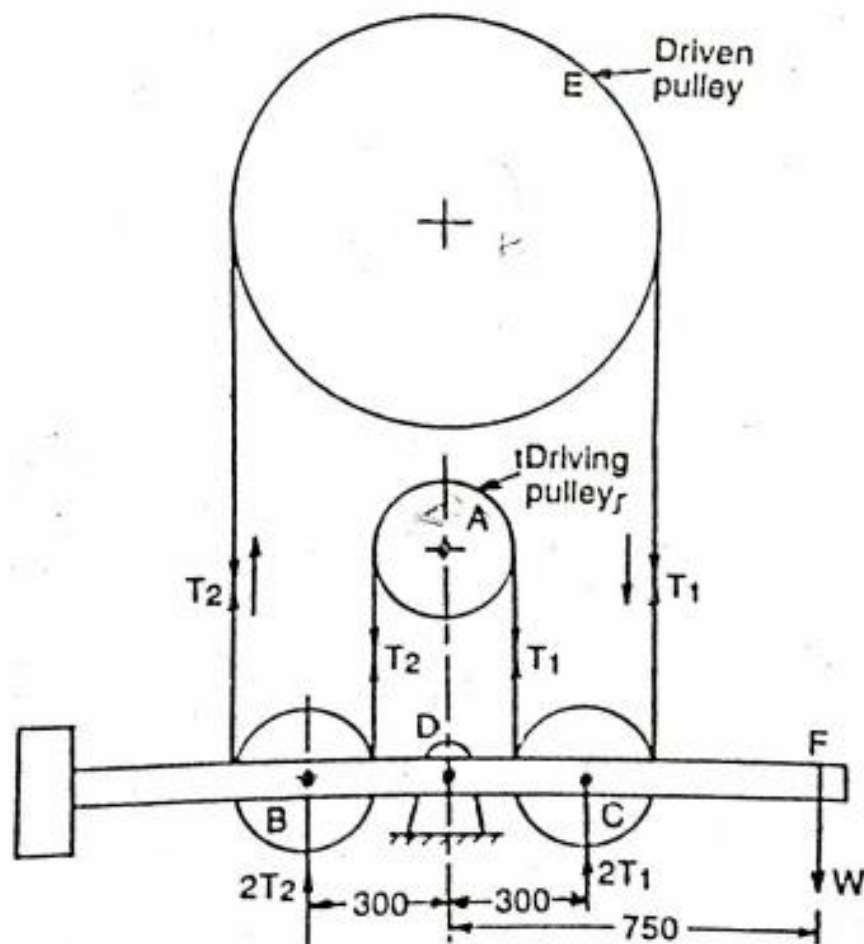


Fig. 19.35 All dimensions in mm.

Fig. 17.55 All dimensions in mm.

Find : 1. the value of the weight W to maintain the beam in a horizontal position when 4.5 kW is being transmitted, and 2. the value of W , when the belt just begins to slip on pulley A. The coefficient of friction being 0.2 and maximum tension in the belt 1.5 kN.

Solution. Given : $N_A = 600$ r.p.m. ; $D_A = D_B = D_C = 300$ mm = 0.3 m

1. *Value of the weight W to maintain the beam in a horizontal position*

Given : Power transmitted (P) = 4.5 kW = 4500 W

Let T_1 = Tension in the tight side of the belt on pulley A, and

T_2 = Tension in the slack side of the belt on pulley A.

$\therefore$ Force acting upwards on the pulley C = $2 T_1$

and force acting upwards on the pulley B = $2 T_2$

Now taking moments about the pivot D,

$$W \times 750 = 2 T_1 \times 300 - 2 T_2 \times 300 = 600 (T_1 - T_2)$$

$$\therefore T_1 - T_2 = W \times 750 / 600 = 1.25 W \text{ N}$$

We know that the power transmitted (P),

$$4500 = \frac{(T_1 - T_2) \pi D_A N_A}{60} = \frac{1.25 \text{ W} \times \pi \times 0.3 \times 600}{60} = 11.78 \text{ W}$$

$$\therefore W = 4500/11.78 = 382 \text{ N Ans.}$$

2. Value of W , when the belt just begins to slip on A

Given : $\mu = 0.2$; $T_1 = 1.5 \text{ kN} = 1500 \text{ N}$

We know that

$$2.3 \log \left(\frac{T_1}{T_2} \right) = \mu \theta = 0.2 \times \pi = 0.6284 \quad \dots (\because \theta = 180^\circ = \pi \text{ rad})$$

$$\log \left(\frac{T_1}{T_2} \right) = \frac{0.6284}{2.3} = 0.2732 \quad \text{or} \quad \frac{T_1}{T_2} = 1.876 \quad \dots (\text{Taking antilog of } 0.2732)$$

$$\therefore T_2 = T_1 / 1.876 = 1500/1.876 = 800 \text{ N}$$

Now taking moments about the pivot D ,

$$\begin{aligned} W \times 750 &= 2 T_1 \times 300 - 2 T_2 \times 300 = 2 \times 1500 \times 300 - 2 \times 800 \times 300 \\ &= 420 \times 10^3 \end{aligned}$$

$$\therefore W = 420 \times 10^3 / 750 = 560 \text{ N Ans.}$$

...used in the dynamometer for accurate measurement.

Example 19.19. A torsion dynamometer is fitted to a propeller shaft of a marine engine. It is found that the shaft twists 2° in a length of 20 metres at 120 r.p.m. If the shaft is hollow with 400 mm external diameter and 300 mm internal diameter, find the power of the engine. Take modulus of rigidity for the shaft material as 80 GPa.

Solution. Given : $\theta = 2^\circ = 2 \times \pi / 180 = 0.035 \text{ rad}$; $l = 20 \text{ m}$; $N = 120 \text{ r.p.m.}$; $D = 400 \text{ mm} = 0.4 \text{ m}$; $d = 300 \text{ mm} = 0.3 \text{ m}$; $C = 80 \text{ GPa} = 80 \times 10^9 \text{ N/m}^2$

We know that polar moment of inertia of the shaft,

$$J = \frac{\pi}{32} (D^4 - d^4) = \frac{\pi}{32} [(0.4)^4 - (0.3)^4] = 0.0017 \text{ m}^4$$

and torque applied to the shaft,

$$T = \frac{C \cdot J}{l} \times \theta = \frac{80 \times 10^9 \times 0.0017}{20} \times 0.035 = 238 \times 10^3 \text{ N-m}$$

We know that power of the engine,

$$P = \frac{T \times 2\pi N}{60} = \frac{238 \times 10^3 \times 2\pi \times 120}{60} = 2990 \times 10^3 \text{ W} = 2990 \text{ kW} \quad \text{Ans.}$$

Thank You